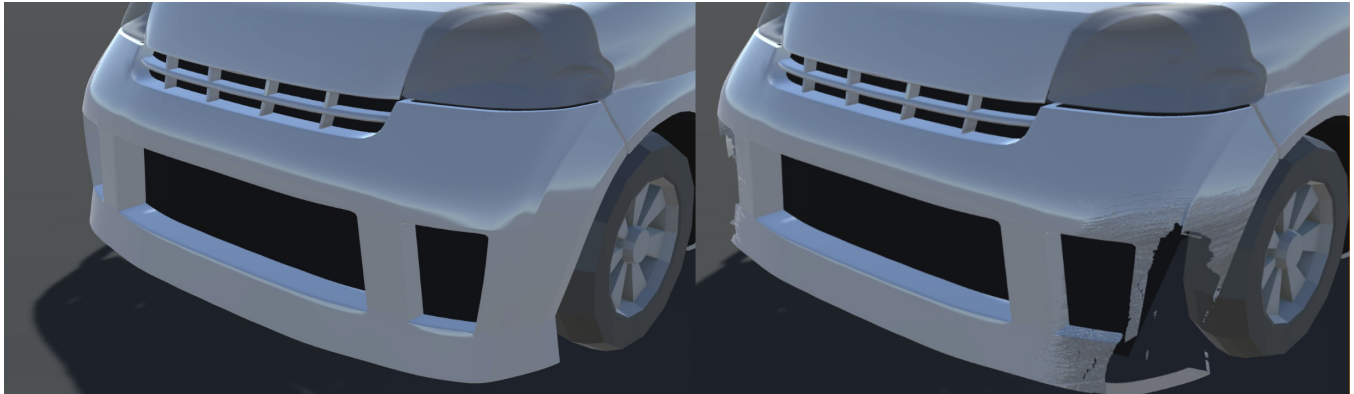


# Representation of FRP material damage in 3DCG

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**Figure 1: Expression of FRP bumper damaged by collision. Left: bumper made of FRP before collision. Right: bumper destroyed after collision.**

## ABSTRACT

In recent years, most productions of 3D video games dealing with cars have represented automobile damage due to collisions. However, the cracks peculiar to FRP materials have not been reproduced in previous works. Therefore, in this research, we devised a method of expressing cracks in FRP materials. In addition, we will conduct experiments to determine the realism of the visual representation of breakage as well as the processing cost and consider whether it is possible to incorporate it into actual video game works.

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## 1 INTRODUCTION

There are many video game productions that are produced using 3DCG, and the realism is improving every year due to technological advances. In recent years, most productions dealing with automobiles have used car damage representations caused by collisions. In this study, we focused on the expression of damage to FRP material parts in works dealing with automobiles. On the other hand, existing works have not reproduced the cracking damage that is

characteristic of FRP materials, and there is no difference in the expression of damage between FRP and exterior parts made of steel plates, with some parts showing metallic luster where the paint has peeled off and others showing the same plastic deformation as steel plates. There are some high-precision and low-cost methods for plastic deformation, such as Jones et al.'s method [Jones et al. 2016]. However, we did not find anything on cracking of FRP materials. Although a method by Kitani et al. [Kitani et al. 2008] exists for breaking paper, which is close as a material with solidified fibers, we thought it would be difficult to apply it to a 3D model on a game work. Since we believe that the realization of FRP material breakage representation will make the in-game experience more realistic and enhance player immersion, we attempted to reproduce FRP material breakage in a game work in a realistic and low-processing-cost manner.

## 2 PROPOSED METHOD

In the proposed method for expressing damage to FRP materials, the conditions were that the method must be easy to implement and be able to run on a variety of game engines. In addition, the implementation of the method on a game must be cost-effective, have no significant impact on the frame rate, and not detract from the user experience. A simple method to reproduce cracks would be to actually change the shape of the model. However, the model used in game productions has a limited number of polygons, making it difficult to achieve detailed expressions, and the computational cost of deformation is expected to increase.

Our method reproduces cracks without changing the shape of the model itself by applying a transparent texture to the cracks. The flow of the proposed method is shown in Figure 2. The shape

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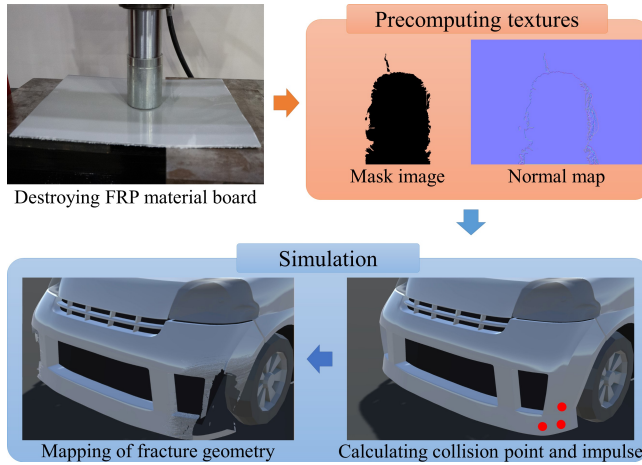


Figure 2: Processing flow of the proposed method.

of the cracks is obtained by generating a texture from the actual FRP material board cracked by applying pressure with a hydraulic press machine. We cracked 38 boards by varying the pressure from 10 MPa to 79 MPa, and calculate the mask and normal maps from photographs of the crack shape, respectively.

The pattern of cracking during the simulation is determined based on impulse obtained at the moment of collision. In this study, scratch-only is applied when the impulse is impacted by less than 2 N·s, small cracking is applied when the impulse is less than 450 N·s, and a larger crack is applied in cases where it was greater than that.

### 3 EXPERIMENTS

We implement our method on Unity. In applying the physics calculations, the weight of the car was set to be equivalent to that of the actual car modeled. Five scenes were prepared for the experiment, as shown in Table 1. The purpose of scene(a), (b), (c), and (d) is to confirm the damage pattern according to the force product, while the purpose of (e) is to confirm the performance and operation of the system in a crash with multiple units. Experiments will be conducted in the game view within the Unity Editor to evaluate corruption representation and to measure processing time when running at 1920 px x 1080 px. The visual aspect of the evaluation is determined visually based on the results of the drawing in the game view. For the evaluation of processing time, the total CPU

Table 1: Conditions and average processing time of the experimental scene.

| Scene | Speed (km/h) | Angle (deg) | # cars | Average processing time (ms) |
|-------|--------------|-------------|--------|------------------------------|
| (a)   | 40           | 90          | 1      | 13.664                       |
| (b)   | 20           | 90          | 1      | 13.086                       |
| (c)   | 40           | 30          | 1      | 11.564                       |
| (d)   | 20           | 30          | 1      | 8.893                        |
| (e)   | 40           | 90          | 16     | 13.143                       |

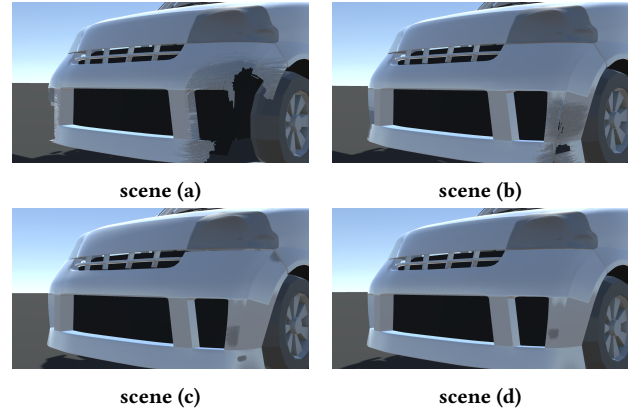


Figure 3: Results for each scene.

processing time for the five drawing scripts is recorded at the time of collision, and the data from a total of five experiments is used to evaluate whether 60 fps can be maintained.

Comparing Figure 3 (a) and Figure 3 (b), it can be seen that the size of the cracks and the size of the scratches change due to the difference in speed. In Figure 3 (c) and Figure 3 (d), the impact is at an angle of 30° to the wall at an angle of 30°, so no cracking occurs, only scratches. We can see that the size of the wound changes with speed. In each scene, there is almost no variation in the breakage pattern, and the conditional branching is functioning properly. Table 1 shows the total processing time for the five rendering scripts. The computation times for each scene were all less than 16.67 ms. In addition, the processing time was considered to be within a practicable range, since the visual frame rate reduction was almost imperceptible.

### 4 SUMMARY AND FUTURE ISSUES

In this study, we verified whether it is possible to express the damage of FRP parts used in automobiles by using texture transparency in a game work using 3DCG. We confirmed that the proposed method can represent the cracking characteristic of FRP material at a relatively low computational cost. In addition, the creation of fracture patterns based on photographs of actual FRP board fractures made it possible to reproduce realistic crack shapes.

On the other hand, future issues include improving the reproduction of glass fibers using normal maps, stabilizing the frame rate by integrating scripts, and reproducing fine cracks by studying the fracture method of FRP plates.

### ACKNOWLEDGMENTS

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